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### **Task Description – 1 (Air Quality API Integration):**

#### **Prompt:**

Create a Python program using the requests library to get real-time air quality data for a given city. Display AQI value, pollution level, and main pollutant in a clear format. Add error handling for invalid city name, wrong API key, and network issues, and keep the code simple.

#### **Code:**

```

# Task Description - 1 (Air Quality API Integration)

import requests

# (function) def get_air_quality(city: Any) -> None
def get_air_quality(city):
    try:
        # Dummy response (simulate API)
        data = {
            "status": "success",
            "data": {
                "city": city,
                "aqi": 120,
                "main_pollutant": "PM2.5"
            }
        }
        if data["status"] == "success":
            aqi = data["data"]["aqi"]
            main_pollutant = data["data"]["main_pollutant"]
            # Pollution level logic
            if aqi <= 50:
                level = "Good"
            elif aqi <= 100:
                level = "Moderate"
            elif aqi <= 150:
                level = "Unhealthy for Sensitive Groups"
            else:
                level = "Unhealthy"
            print("\nAir Quality Details")
            print("City:", city)
            print("AQI:", aqi)
            print("Pollution Level:", level)
            print("Main Pollutant:", main_pollutant)
        else:
            print("Error: City not found")
    except Exception:
        print("Error: Something went wrong")

if __name__ == "__main__":
    city = input("Enter city: ")
    get_air_quality(city)

```

**Output:**

```
● PS C:\Users\sampe> & 'c:\Users\sampe\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\sampe\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '55370' '--' 'C:\Users\sampe\Ai Assignment Coding Assisgnment 18.2(1453)(1).py'
Enter city: delhi

Air Quality Details
City: delhi
AQI: 120
Pollution Level: Unhealthy for Sensitive Groups
Main Pollutant: PM2.5
○ PS C:\Users\sampe> 
```

## Task Description – 2 (Language Translation API):

**Prompt:** Create a Python program using requests library to translate text from English to two languages. Validate user input and language codes, and handle errors like invalid input or API failure. Display original text and translated text clearly, keeping the code simple.

**Code:**

```

1  # Task Description - 2 (not explicitly named, but appears to be translation)
2
3  import requests
4
5  def translate_text(text, lang):
6      # Dummy translations
7      translations = {
8          "fr": "Bienvenue dans le codage assisté par IA",
9          "es": "Bienvenido a la codificación asistida por IA",
10         "hi": "एआई आधारित कोडिंग में आपका स्वागत है"
11     }
12     try:
13         if not text.strip():
14             print("Error: Please enter valid text")
15             return
16         if lang not in translations:
17             print("Error: Invalid language code")
18             return
19         print("\nOriginal Text:", text)
20         print("Translated Text:", translations[lang])
21     except Exception:
22         print("Error: Translation service unavailable")
23
24     # User input
25     text = input("Enter text:")
26     lang1 = input("Enter first language (fr/es/hi):")
27     lang2 = input("Enter second language (fr/es/hi):")
28
29     translate_text(text, lang1)
30     translate_text(text, lang2)

```

**Output:**

```

PS C:\Users\sampe> c:: cd 'c:\Users\sampe'; & 'c:\Users\sampe\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\sampe\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58746' '--' 'C:\Users\sampe\Ai Assignment Coding Assisgnment 18.2(1453)(1).py'
Enter text:Welcome to AI Assignment Coding
Enter first language (fr/es/hi):fr
Enter second language (fr/es/hi):es

Original Text: Welcome to AI Assignment Coding
Translated Text: Bienvenue dans le codage assisté par IA

Original Text: Welcome to AI Assignment Coding
Translated Text: Bienvenido a la codificación asistida por IA

PS C:\Users\sampe> c:: cd 'c:\Users\sampe'; & 'c:\Users\sampe\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\sampe\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '65481' '--' 'C:\Users\sampe\Ai Assignment Coding Assisgnment 18.2(1453)(1).py'
Enter text:Welcome to AI Assignment Coding
Enter first language (fr/es/hi):hi
Enter second language (fr/es/hi):fr

Original Text: Welcome to AI Assignment Coding
Translated Text: आई आधारीत कोरिडि में अप्प स्वागत है

Original Text: Welcome to AI Assignment Coding
Translated Text: Bienvenue dans le codage assisté par IA

```

### Task Description – 3 (Public Book Information API):

#### Prompt:

Create a **Python program using requests library** to fetch book details from a public API using book title or ISBN.

Display **title, author, publisher, and published year** clearly.

Add **error handling for invalid input, book not found, and network issues**, and keep the code simple.

#### Code:

```

1  # Task Description - 3 (Public Book Information API)
2
3  import requests
4
5  def get_book_info(query):
6      try:
7          if not query.strip():
8              print("Error: Please enter a valid book title")
9              return
10         url = f"https://openlibrary.org/search.json?q={query}"
11         response = requests.get(url, timeout=5)
12         data = response.json()
13         if data['numFound'] == 0:
14             print("Error: Book not found")
15             return
16         book = data['docs'][0]
17         title = book.get('title', 'N/A')
18         author = book.get('author_name', ['N/A'])[0]
19         publisher = book.get('publisher', ['N/A'])[0]
20         year = book.get('first_publish_year', 'N/A')
21         print("\nBook Details")
22         print("Title:", title)
23         print("Author:", author)
24
25         print("Publisher:", publisher)
26         print("Year:", year)
27     except requests.exceptions.RequestException:
28         print("Error: Network issue or API unavailable")
29     except Exception as e:
30         print("Error:", e)
31
32 # User input
33 query = input("Enter book title: ")
get_book_info(query)

```

**Output:**

```
3\python.exe' 'c:\Users\sampe\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '65481' '--' 'C:\Users\sampe\Ai Assignment Coding Assisgnment 18.2(1453)(1).py'
PS C:\Users\sampe> c;; cd 'c:\Users\sampe'; & 'c:\Users\sampe\AppData\Local\Programs\Python\Python31
3\python.exe' 'c:\Users\sampe\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '49876' '--' 'C:\Users\sampe\Ai Assignment Coding Assisgnment 18.2(1453)(1).py'
Enter book title: Artificial Intelligence

Book Details
Title: Artificial intelligence
Author: Stuart J. Russell
Publisher: N/A
Year: 1994

PS C:\Users\sampe> c;; cd 'c:\Users\sampe'; & 'c:\Users\sampe\AppData\Local\Programs\Python\Python31
3\python.exe' 'c:\Users\sampe\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57966' '--' 'C:\Users\sampe\Ai Assignment Coding Assisgnment 18.2(1453)(1).py'
Enter book title: Python Programming

Book Details
Title: Core Python Programming
Author: R. Nageswara Rao
Publisher: N/A
Year: 2016
```

## Task Description – 4 (Real-Time Application: Job Listings Aggregator)

### Prompt:

Create a Python program using requests library to fetch job listings from a public Jobs API based on a keyword or location.

Display job title, company name, location, and posting date in a numbered list.

Add error handling and retry mechanism for failed requests, and keep the code simple.

### Code:

```

1  import requests
2  import time
3
4  def get_jobs(keyword):
5      url = f"https://remotive.com/api/remote-jobs?search={keyword}"
6      retries = 3
7
8      for attempt in range(retries):
9          try:
10             response = requests.get(url, timeout=10)
11
12             if response.status_code != 200:
13                 print(f"Attempt {attempt+1}: API returned status {response.status_code}")
14                 if attempt < retries - 1:
15                     time.sleep(2)
16                     continue
17
18             data = response.json()
19             jobs = data.get("jobs", [])
20
21             if not jobs:
22                 print("Error: No jobs found")
23                 return
24
25             print("\nJob Listings:\n")
26             for i, job in enumerate(jobs[:5], start=1):
27                 print(f"{i}. {job.get('title')}")
28                 print(f"    Company: {job.get('company_name')}")
29                 print(f"    Location: {job.get('candidate_required_location')}")
30                 print(f"    Date: {job.get('publication_date')}\n")
31             return
32
33             except requests.exceptions.RequestException:
34                 print(f"Attempt {attempt+1}: Network error, retrying...")
35                 time.sleep(2)
36
37             print("Error: Failed to fetch jobs after multiple attempts")
38
39             # User input
40             keyword = input("Enter job role: ")
41             get_jobs(keyword)

```

**Output:**



```
● PS C:\Users\sampe> & 'c:\Users\sampe\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\sampe\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '61388' '--' 'C:\Users\sampe\Ai Assignment Coding Assignment 18.2(1453)(1).py'
Enter job role: Python developer
Error: No jobs found

● PS C:\Users\sampe> c:; cd 'c:\Users\sampe'; & 'c:\Users\sampe\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\sampe\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '60629' '--' 'C:\Users\sampe\Ai Assignment Coding Assignment 18.2(1453)(1).py'
Enter job role: Data analyst

Job Listings:

1. Online Data Analyst United States
   Company: TELUS Digital
   Location: USA

2. Online Data Analyst (United States/Spanish speakers)
   Company: TELUS Digital
   Location: USA
   Date: 2026-04-07T14:40:01
```

## Task Description – 5 Stock Market Data API Integration

### Prompt:

Create a **Python program using requests library** to fetch stock market data for a given company symbol.

Display **stock name, current price, day high, day low, and percentage change** clearly.

Add **input validation and error handling for invalid symbol, API issues, and network errors**, and keep the code simple.

### Code:

```

1  # Task Description - 5 (Stock Market Data API Integration)
2
3  import requests
4
5  def get_stock_data(symbol):
6      try:
7          if not symbol.strip():
8              print("Error: Please enter a valid stock symbol")
9              return
10         # Dummy stock data (simulation)
11         stock_data = {
12             "AAPL": {
13                 "name": "Apple Inc",
14                 "price": 175.5,
15                 "high": 180.8,
16                 "low": 170.2,
17                 "change": "+1.2%"
18             },
19             "TSLA": {
20                 "name": "Tesla Inc",
21                 "price": 250.3,
22                 "high": 255.0,
23                 "low": 245.5,
24             },
25         }
26         symbol = symbol.upper()
27         if symbol not in stock_data:
28             print("Error: Invalid stock symbol")
29             return
30         stock = stock_data[symbol]
31         print("\nStock Details")
32         print("Company:", stock["name"])
33         print("Current Price:", stock["price"])
34         print("Day High:", stock["high"])
35         print("Day Low:", stock["low"])
36         print("Change:", stock["change"])
37     except Exception:
38         print("Error: Network issue or API failure")
39
40 # User input
41 symbol = input("Enter stock symbol: ")
42 get_stock_data(symbol)

```

**Output:**

```
PS C:\Users\sampe> c::; cd 'c:\Users\sampe'; & 'c:\Users\sampe\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\sampe\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '61773' '--' 'C:\Users\sampe\Ai Assignment Coding Assisgnment 18.2(1453)(1).py'  
Enter stock symbol: AAPL
```

```
Stock Details  
Company: Apple Inc  
Current Price: 175.5  
Day High: 180.8  
Day Low: 170.2  
Change: +1.2%
```

```
PS C:\Users\sampe> c::; cd 'c:\Users\sampe'; & 'c:\Users\sampe\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\sampe\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '60097' '--' 'C:\Users\sampe\Ai Assignment Coding Assisgnment 18.2(1453)(1).py'  
Enter stock symbol: TSLA
```

```
Stock Details  
Company: Tesla Inc  
Current Price: 250.3  
Day High: 255.0  
Day Low: 245.5  
Change: -0.8%
```